

SALMAN SHAIKH

Python Developer

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SUMMARY

- Backend Engineer specializing in distributed systems, real-time communication, serverless pipelines, and large-scale data processing. Experienced with FastAPI, Django, Elasticsearch, MongoDB, SQL, and cloud platforms (AWS, GCP). Built microservices handling terabytes of data, optimized algorithms for high-volume workloads, and delivered production-grade systems for real-time chat, voice calling, ETL pipelines, AI assistants, and LLM fine-tuning.

EXPERIENCE

Software Engineer

Mumbai, India

Houzeo

08/2024 - Present

- Architected and developed a WebSocket-based communication microservice enabling real-time in-app chat and voice calling across iOS and Android using FastAPI.
- Integrated Agora SDKs and webhooks for secure token generation, voice call recording, and reliable real-time communication flows.
- Designed and implemented Redis Pub/Sub for scalable inter-instance communication across distributed FastAPI servers.
- Integrated Elasticsearch as the primary data store for chat and call logs, enabling fast message indexing, search, and analytics.
- Built and maintained SQL persistence layers for user management, permissions, and transactional workflows.
- Implemented push notification systems for chat and VoIP calls using Firebase Cloud Messaging (FCM) and Apple Push Notification Service (APNs).
- Led research and POC development for the entire communications module, validating architecture, scaling strategies, and production readiness.
- Deployed and monitored microservices on AWS, implementing autoscaling, centralized logging, health checks, and performance monitoring for high-availability communication services.

Software Engineer

Remote

RaftLabs

12/2023 - 06/2024

- Developed and maintained cross-platform Flutter applications using GetX for efficient state management and reactive UI updates.
- Integrated GraphQL APIs within Flutter apps to enable optimized data querying and real-time synchronization between client and server.
- Implemented push notification systems for real-time updates and user engagement using Firebase Cloud Messaging (FCM).
- Built and integrated Agora-based audio conferencing rooms, ensuring seamless multi-user communication and low-latency performance.
- Collaborated on the development of an admin dashboard built with React, contributing to UI enhancements, API integrations, and overall feature optimization.
- Managed full-stack integration between Flutter (frontend), FastAPI/GraphQL (backend), and cloud infrastructure, ensuring high scalability and maintainability across modules.

Software Engineer

Location

VirtouStack Softwares Pvt Ltd

Date period

- Led a team of two developers to build a scalable serverless ETL pipeline using GCP/Firebase Functions, Firestore triggers, and Cloud Scheduler for logistics clients like Tata and Nestlé.
- Managed multi-environment deployments, log monitoring, and provided technical support to client engineering teams.
- Implemented batch-processing workflows capable of handling thousands of records per request, ensuring stable and high-performance data processing across the ETL system.
- Enforced strict data typing and schema validation in all ETL stages using Python-based serverless functions, increasing overall data accuracy to 99%.
- Designed complete database schemas and data-flow architectures for multiple large-scale applications, ensuring long-term scalability, optimized reads/writes, and clean relationships across NoSQL structures.
- Developed REST APIs and webhook integrations for third-party services including Razorpay, Algolia, and others, enabling secure payments, search indexing, and external workflow automations.
- Built serverless functions for PDF generation and storage in Google Cloud Storage, automating documentation, invoicing, and transactional workflows.
- Engineered a real-time driver assignment algorithm using GeoFire geospatial queries in Firestore, optimizing driver matching, lowering ETA, and improving delivery efficiency.
- Developed and deployed authentication systems for mobile apps using Django, hosted on AWS EC2, and managed the complete microservice architecture for performance and reliability.
- Scheduled and maintained cron jobs for FCM notifications, subscription renewals, reporting workflows, and other automated backend operations.
- Built analytics and monitoring modules to track ETL accuracy, system uptime, and failure patterns, and integrated automated email reporting for daily summaries and crash alerts.
- Collaborated closely with frontend/mobile teams to ensure seamless integration between backend logic, Flutter apps, and third-party services across DigiCourier and Humsafer platforms.

PROJECTS

Reacher

06/2025
Mumbai, India

Python, Serverless Functions, Gemini API, Gmail API, Outlook API, Google Auth

- Integrated Google's Gemini API to generate AI-powered email templates, enabling users to create personalized and context-aware content instantly.
- Implemented Gmail API integration for sending emails directly from the application, including Google OAuth with advanced mail-sending scopes.
- Added Outlook authentication and mail-sending workflows, ensuring seamless support for both major email providers.
- Built backend logic using Python and serverless functions, creating scalable and low-latency email automation pipelines.
- Developed a batch email processing system capable of sending messages to all stored contacts with 99% delivery reliability and minimal latency.
- Ensured robust security and compliance by managing OAuth permissions, access tokens, and secure API interactions across Google and Microsoft services.

LAAS (LLM-as-a-Service)

2025
Mumbai, India

Python, Pandas, Gemma 270M, PEFT LoRA, Transformers, NVIDIA Accelerated Training

- Designed a niche-based LLM system (SLMs) that delivers domain-specific responses with 40% higher accuracy while reducing compute cost by up to 80%, resulting in a faster, cheaper, and more precise AI model.
- Preprocessed and transformed large CSV datasets using Pandas, converting raw data into optimized, LLM-ready training formats tailored for each niche.
- Fine-tuned Google's Gemma 270M model using PEFT LoRA, leveraging HuggingFace Transformers and custom tokenizers to build high-efficiency specialized models.
- Implemented checkpoint-based fine-tuning, enabling recoverable, iterative, and stable training cycles for long-running jobs.
- Utilized NVIDIA GPU-accelerated training pipelines in Python, optimizing throughput and reducing fine-tuning time across all niche LLM variants.
- Developed a scalable training workflow that supports continuous model updates, easy deployment, and automated benchmarking for accuracy and cost efficiency.

Remindel

08/2025
Mumbai, India

AI-Powered Voice Reminder System

- Built an AI-driven reminder app that converts calendar data into meaningful, language-specific voice messages delivered via automated phone calls.
- Used Gemini API to generate personalized reminder scripts in multiple languages based on user calendar entries.
- Integrated Google Text-to-Speech to convert AI-generated content into natural-sounding audio messages.
- Added multilingual support (Hindi, Marathi, Malayalam, etc.) with dynamic language switching.
- Created a scheduled cron-based calling system that triggers call notifications exactly at reminder time.
- Deployed the backend using Google serverless functions and managed scheduled execution using cloud-based cron triggers.

EDUCATION

Bachelor of engineering from North Maharashtra University

Jalgaon, India
06/2017 - 08/2021

Raisoni Group of Institutions

- Computer Science Engineering
- **Mini Project — Windows PDF Analytics Application**
- Developed a Windows-based PDF analytics tool using Python (Tkinter, PyPDF, NumPy, and Pandas) to extract, analyze, and visualize structured data from PDF documents.
- **Major Project — Malicious User Detection Using Honeywords**
- Built a malicious-user detection system leveraging the Honeywords technique, implemented with Python, FastAPI, and SQL. Designed secure authentication workflows, decoy password generation, and threat-detection logic to identify unauthorized access attempts in real time.

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